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# Plastic pollution in agrifood systems and One Health

Impact pathways and policy implications





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## About this brief

This technical brief was developed following the workshop “Plastic Pollution and One Health: Understanding and Managing Risks in Agrifood Systems”, convened at the FAO headquarters in Rome, Italy, on 18–19 November 2025.

The workshop was jointly organized by FAO, the Agriculture, Nutrition & Health (ANH) Academy Science-Policy Platform of the London School of Hygiene & Tropical Medicine (LSHTM), and Università Cattolica del Sacro Cuore (UCSC). It brought together experts from diverse disciplines, sectors and regions to examine the implications of plastic pollution in agrifood systems through a One Health lens.

This work was co-led by LSHTM and FAO who developed the overall conceptual approach, evidence mapping and preliminary framework. Through the workshop and subsequent virtual consultations, participants co-developed evidence-based impact pathways to better understand the cascading impacts of plastic pollution across One Health domains. This technical brief draws on the evidence, discussions and expert contributions generated through that process.

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# Executive summary

Plastic pollution is pervasive across agrifood systems and presents a major and growing global concern for people and the planet. Plastics are embedded throughout agrifood value chains, from agricultural production to food processing, packaging, transport and waste management, which contribute directly to agrifood system pollution and food contamination. Agrifood systems are also exposed to plastic pollution originating in other sectors, transported through wastewater, runoff and atmospheric deposition or introduced through contaminated irrigation water and soil amendments. Across all sectors and throughout their life cycles, plastics generate waste and debris, micro- and nanoplastics (MNPs), hazardous chemicals and air pollutants, and contribute to greenhouse gas emissions. These forms of pollution contaminate agrifood systems and can disrupt the environmental foundations of food production, generating cascading harms across interconnected systems.

One Health is an integrated, unifying approach that aims to sustainably balance and optimize the health of humans, animals, plants and ecosystems. Indigenous and traditional knowledge systems have long understood health as inseparable from the well-being of the lands, waters and living beings that sustain communities. The growing policy and scientific framing of One Health recognizes that the health across human, animal, plant and environmental domains is closely linked and interdependent, and mobilizes sectors, disciplines and communities to work together to address threats to health and ecosystems and contribute to sustainable development.

In the context of plastic pollution, One Health provides a critical approach for understanding the interconnected pathways of exposure and impacts across these domains and for designing effective responses. Applying a One Health lens can connect evidence across sectors and disciplines, identify positive and negative feedback loops and trade-offs, and support more coordinated, preventive and transformative action across agrifood systems.

This technical brief describes four illustrative impact pathways that trace evidence of the cascading effects of plastic pollution in agrifood systems on One Health. Together, these pathways highlight examples of cross-cutting and compounding threats for agrifood systems, and across environments, plants, animals, human and planetary health and well-being.

**1. Cascading impacts through soils.** Plastic pollution is accumulating rapidly in soils, where MNPs can disrupt soil structure, microbiota, nutrient and carbon cycling, and introduce or mobilize chemical and microbial hazards. MNPs can impair plant growth and productivity and be taken up into crops, creating potential exposure pathways through food and feed. Together, these effects threaten climate, ecosystem services, agricultural productivity, and food security, with cascading impacts across One Health.

**2. Animals at risk across land and water.** Plastic ingestion and entanglement widely affects aquatic and terrestrial animals, causing injury, impaired nutrition and growth, disease and mortality. Plastic particles, their associated chemicals, and the wider pollutants, microbes and pathogens they transport can also move through food webs and animal products, with potential implications for food safety and antimicrobial resistance. Together, these impacts have cascading consequences for livelihoods, food security and nutrition, particularly for communities highly dependent on fisheries, livestock and working animals.

**3. Human exposure and system feedbacks.** Humans are exposed throughout life to plastic chemicals and MNPs through dietary and environmental contamination, handling food-contact materials, and occupational exposures across the agrifood system. Plastic-associated chemicals are linked to endocrine disruption, reproductive and developmental effects, while emerging evidence suggests MNPs may contribute to inflammation, microbiome disruption, tissue damage and reproductive toxicity. These pollutants can recirculate through wastewater and agricultural soils, creating feedback loops that sustain exposure and risks across humans, animals and ecosystems.

**4. Climate and nature under pressure.** Pollution and emissions released across plastics life cycles contribute to the triple planetary crisis of climate change, pollution and biodiversity loss. MNPs can transport hazardous chemicals and provide surfaces for microbial colonization, potentially increasing risks from pathogens, antimicrobial resistance and food contamination. Impacts on pollinators, beneficial species and other organisms further threaten ecosystem services essential to food production. Together, these systemic effects create cascading risks for agrifood systems and the health of humans, animals, plants and the environment.

Translating this evidence into action requires responses across sectors, the plastics life cycle and agrifood systems as a whole. Drawing on the ongoing work towards the FAO One Health in Agrifood Systems Framework, this brief identifies four mutually reinforcing areas for action.<sup>1</sup> First, **a shared evidence base connected across sectors**. Underpinned by harmonized and interoperable surveillance, monitoring and assessment systems, integrated data governance and science-policy interfaces, to advance a shared understanding of plastics' associated hazards across soil, water, biota, food webs and human health systems. Second, **inclusive solutions co-designed with communities** should engage and facilitate meaningful participation of affected groups, including smallholders, waste pickers, Indigenous Peoples, artisanal fishers and women's groups, to ensure measures are equitable, feasible and do not create new harms. Third, **coordinated action across policies and global agreements** requires coordination across ministries and alignment with relevant multilateral agreements, including the emerging global plastics treaty and biodiversity and climate frameworks. Fourth, **responsive review informed by living evidence**, supported by strengthened international expertise and community-engaged monitoring to inform regular policy review and horizon scanning by multi-stakeholder bodies protected from conflicts of interest.

The evidence shows that responses to pollution that focus only on downstream plastic waste management will not be sufficient, and can result in shifting the burdens of pollution rather than reducing them at source. In this context, this brief supports a just transition toward more sustainable, efficient, resilient and equitable agrifood systems through the prevention and reduction of plastic pollution across the full life cycle. This transition, informed by a One Health approach, should be guided by the precautionary principle, inclusive governance and careful assessment of alternatives, to ensure that replacement materials, technologies and practices are safe, effective and do not introduce new risks to food security and safety, environmental integrity, or animal and human health.



<sup>1</sup> The FAO One Health in Agrifood Systems Framework is currently under development, in response to recommendations of the 29th Session of the FAO Committee on Agriculture (COAG). See FAO. 2024. *Report of the 29th Session of the Committee on Agriculture, Rome, 30 September–4 October 2024*. Rome, FAO. <https://openknowledge.fao.org/server/api/core/bitstreams/1a8ee24e-6b85-4831-8a11-8515493cacdc/content>

# 1. The pervasive nature of plastic pollution in agrifood systems

Plastic pollution, including that associated with agrifood systems, is a defining global challenge of our time with far-reaching consequences for people and planet. Plastics are embedded throughout agrifood value chains, from agricultural films, mulches, greenhouse covers, coated fertilizers and silage wraps, to storage containers, processing equipment, food packaging and waste management materials (Duquesne et al., 2025; Hofmann et al., 2023; Yates et al., 2021). Many of these plastics are single-use and short-lived, particularly agricultural products and packaging, most of which becomes waste in less than a single year (Geyer et al 2017; FAO, 2021). The vast majority end up in landfills, are openly burned, or pollute the natural environment, including the lands, waterways and marine environments where food is sourced and produced. Agrifood systems also suffer from broader sources of plastic pollution, transferred across environments, including in runoff from roads, wastewater

flows, atmospheric deposition, and through agricultural applications of contaminated irrigation water and soil amendments (Hooge et al., 2024; Thompson et al., 2024; FAO, 2021).

The scale of agrifood systems' dependence on plastics remains substantial. An estimated 13.4 million tonnes of agricultural plastics are used annually (Nizzetto et al., 2025), alongside 37.3 million tonnes in food packaging (FAO, 2021). Demand for plastics is projected to increase further in the coming decades, with global use of agricultural plastics expected to rise to 19 million tonnes by 2040, while packaging demand is projected to increase by a further 38 percent over the same period (Pew Charitable Trusts, 2025). Rising production and consumption of plastics compound pressures on already overburdened waste management systems and drives mounting levels of environmental pollution (OECD, 2022; Pew Charitable Trusts, 2025; Cowger et al., 2024).



Plastic pollution is not just waste. Plastics are a diverse group of chemically complex materials, with varied physical and functional properties ([Duquesne et al., 2025](#)). Seven major types of plastics are currently produced and consumed in large quantities, all of which are produced from fossil fuel feedstocks.<sup>2</sup> Plastics life cycles span the extraction and processing of raw materials, design, manufacturing, transportation, packaging, distribution, use, reuse, recycling and their various end-of-life fates ([UNEP, 2023b](#)). At each stage, harmful emissions and pollution are released including greenhouse gases (GHGs), air pollutants, toxic chemicals, plastic waste and litter, and MNPs ([Deeney et al., 2026](#); [Thompson et al., 2024](#); [Landrigan et al., 2025](#)). In turn, these emissions and forms of pollution pose serious risks to agrifood systems functioning ([Yates et al., 2025](#)) and to broader environmental, plant, animal and human health ([Scientists' Coalition for an Effective Plastics Treaty, 2026](#)).

GHGs are emitted across the plastics life cycle, contributing directly to the climate crisis. The majority of these emissions come from plastics production processes, beginning with the extraction of fossil fuel feedstocks ([Karali et al., 2024](#); [Pew Charitable Trusts, 2025](#)). Rising temperatures and extreme weather events are a major driver of biodiversity loss and expose millions of people to food insecurity ([Romanello et al., 2025](#)). In 2024, 61 percent of global land area was affected by extreme drought alongside one heat-related human death per minute, on average ([Romanello et al., 2025](#)). Air pollution, including emissions from plastics production and waste burning, remains the leading environmental risk factor for death around the world ([Health Effects Institute, 2025](#)). Plastics contain numerous chemicals, including additives, residual raw materials, processing aids and non-intentionally added substances (NIASs) ([Monclús et al., 2025](#)). A quarter of these chemicals are known to be hazardous, including those with endocrine-disrupting properties, such as bisphenols, phthalates, and certain per- and polyfluoroalkyl substances (PFAS) and flame retardants, and 66 percent of plastics-related chemicals lack the hazard information needed to ascertain their safety ([Monclús et al., 2025](#)).

These chemicals have been detected across all major environmental compartments, including air, freshwater, marine environments, soils, sediments, biota, foods, and in human tissues including documented exposure of infants in utero ([Metcalf et al., 2022](#); [Symeonides et al., 2024](#)).

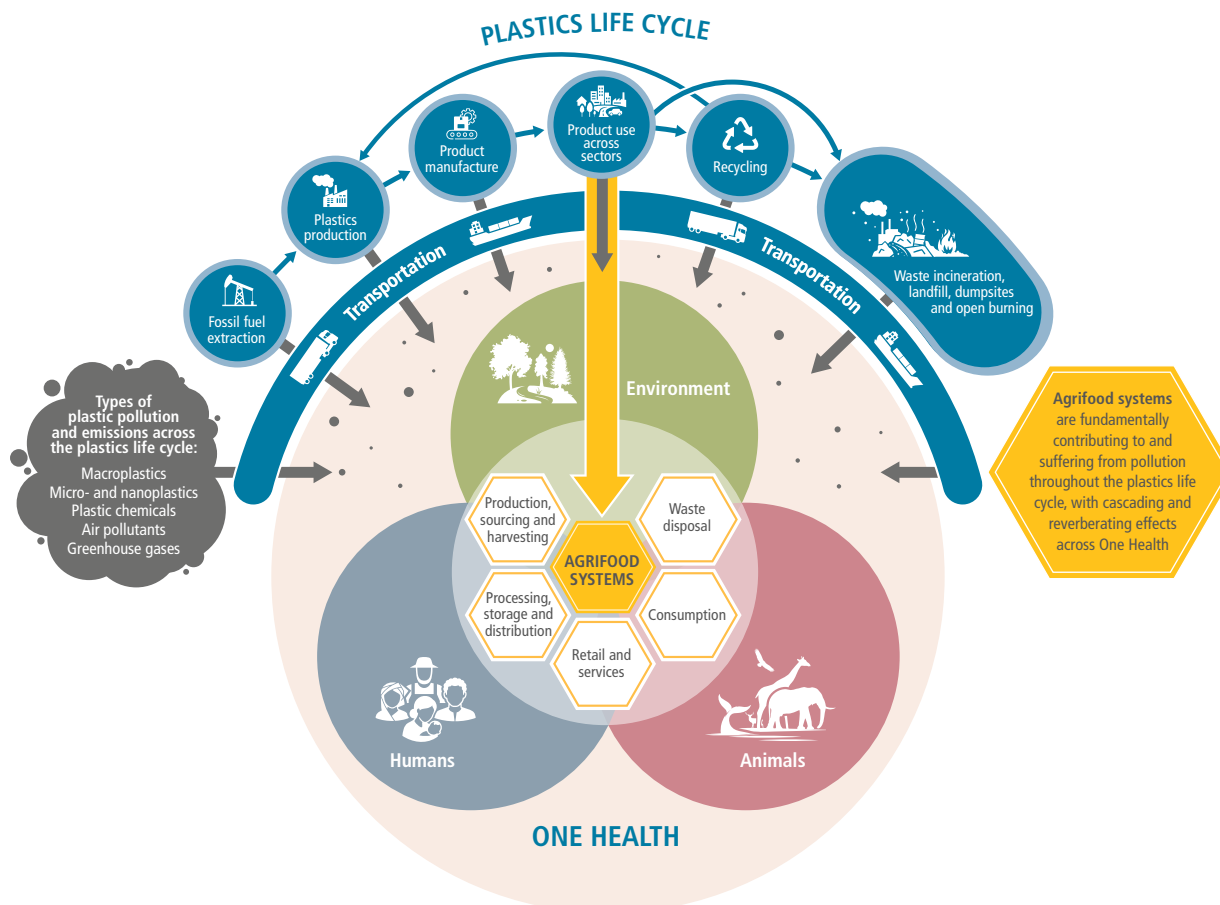
Macroplastic pollution (particles > 5mm) includes plastic objects and materials released to the environment. This form of pollution is associated with risks to food production, sanitation systems, livestock and wildlife, and with the spread of infectious disease and antimicrobial resistance (AMR) ([FAO, 2021](#); [Haddy et al., 2025](#); [Landrigan et al., 2025](#)). Microplastics are solid plastic particles ≤ 5mm in size that are composed of polymers, together with functional additives as well as other intentionally and unintentionally added chemicals, which may be manufactured, or formed and released from larger plastics across their life cycles. Nanoplastics are defined as particles ≤ 1µm ([Thompson et al., 2024](#)). Plastic particles and associated chemicals have been detected across atmospheric, cryospheric, terrestrial and aquatic environments ([Thompson et al., 2024](#)), in drinking water ([Oliveri Conti et al., 2026](#)) and in feed and food commodities, establishing multiple exposure pathways for humans, livestock and wildlife ([FAO, 2022a](#); [Thompson et al., 2024](#); [Nizzetto et al., 2025](#)).

Together, these multiple forms of plastic pollution undermine the resilience and sustainability of agrifood systems, and threaten biodiversity, plant, animal and human health. These impacts translate into substantial societal and economic costs, including from reduced agricultural and fisheries productivity ([FAO, 2021](#)), compromised food safety ([FAO, 2022b](#)), increased food-system and healthcare expenditures, and the degradation of ecosystem services that underpin food security and livelihoods ([FAO, 2021](#); [OECD, 2022](#); [UNEP, 2023b](#); [Duquesne et al., 2025](#)). Deep environmental injustices across the life cycle of plastics mean that Indigenous Peoples and other frontline communities bear the disproportionate burdens of pollution. ([UNEP, 2021b](#); [Orellana, 2021](#)) Collectively, plastic pollution is exacerbating pressures across the Earth's biophysical systems, contributing to compounding and cascading planetary risks for the environment, all forms of life, ecosystems, and human societies ([Villarrubia-Gómez et al., 2024](#)).

<sup>2</sup> These include polyethylene (PE; including LDPE, LLDPE and HDPE), polypropylene (PP), polyethylene terephthalate (PET), polyvinyl chloride (PVC), polystyrene (PS) and related styrenics (e.g. SAN and ABS), and polyurethane (PU) ([Karali et al., 2024](#)).

## 2. A One Health approach to plastic pollution

FIGURE 1  
Plastic pollution and One Health in agrifood systems



Source: Authors' own elaboration

Historically applied to zoonotic diseases and AMR, One Health has evolved to encompass the ecological processes and interconnections that link environmental, plant, animal and human health, including the role of microbiomes (Winkler et al., 2025; FAO, 2026). While One Health has gained prominence as a contemporary policy and scientific framework, many Indigenous knowledge systems have long understood health as inseparable from the well-being of lands, waters, animals, plants and people. Recognizing and centring these knowledge systems is important not only because they offer valuable insights into interconnected environmental and health challenges, but also because Indigenous Peoples and frontline communities are often among those most directly and disproportionately affected by environmental degradation and pollution while contributing least to its drivers (Redvers et al., 2026). Figure 1 presents an overview of the links between plastic pollution, One Health, and agrifood systems.

**One Health** is an integrated, unifying approach that aims to **sustainably balance and optimize the health of humans, animals, plants and ecosystems**. It recognizes that the **health of humans, domestic and wild animals, plants and the wider environment (including ecosystems)** are closely linked and interdependent. The approach mobilizes multiple sectors, disciplines and communities at varying levels of society to work together to foster well-being and tackle threats to health and ecosystems, while addressing the collective need for clean water, energy and air, safe and nutritious food, taking action on climate change, and contributing to sustainable development (OHHLEP, 2022).

One Health offers a coherent and prevention-oriented framework through which to understand and address plastic pollution ([Morrison et al., 2022](#); [OHHLEP, 2022](#); [Wang et al., 2024](#)). Its emphasis on integrated surveillance and transdisciplinary collaboration across environmental, animal, plant and human sectors can support not only the mitigation of cascading impacts, but also their early prevention by enabling observations and monitoring data from one sector to inform hazard identification, risk assessment and timely action in others ([WHO, FAO, UNEP & WOA, 2022](#); [WHO, FAO, UNEP & WOA, 2023](#); [Winkler et al., 2025](#)). This is consistent with the One Health Joint Plan of Action (2022–2026) ([One Health Initiative, 2022](#)) of the Quadripartite organizations (FAO, UNEP, WHO and WOA), which recognize environmental drivers, including pollution, as integral to One Health governance and action.

Negotiations towards an international legally binding instrument on plastic pollution, including in the marine environment, mandated by UNEA resolution 5/14 (hereinafter the “global plastics treaty”), have been underway since 2022. There is growing recognition among Member States of the need for coordinated action across the full life cycle

of plastics. While multilateral negotiations continue, there is also a pressing need for practical, cross-sectoral approaches to better understand, prevent and mitigate the impacts of plastic pollution across agrifood systems.

Applying a One Health approach to plastic pollution in and of agrifood systems enables three mutually reinforcing objectives:

**1. Rigorous characterization** of impact pathways and ecological–health linkages across agrifood system production, processing, distribution and consumption.

**2. Prioritization of interventions** that phase down and transition away from plastic inputs to significantly reduce or eliminate exposure routes while safeguarding food system functions.

**3. Design of governance mechanisms** that mitigate conflicts of interest and incorporate independent scientific evidence, including the knowledge, science, and practices of Indigenous Peoples, with their full and effective participation, with inclusive stakeholder and equity considerations at local, national and international scales ([Jacobs, 2025](#); [Dauvergne, 2023](#)).

#### Explore further: Impact pathways in the interactive [Plastic Pollution in Agrifood Systems and One Health Framework](#)

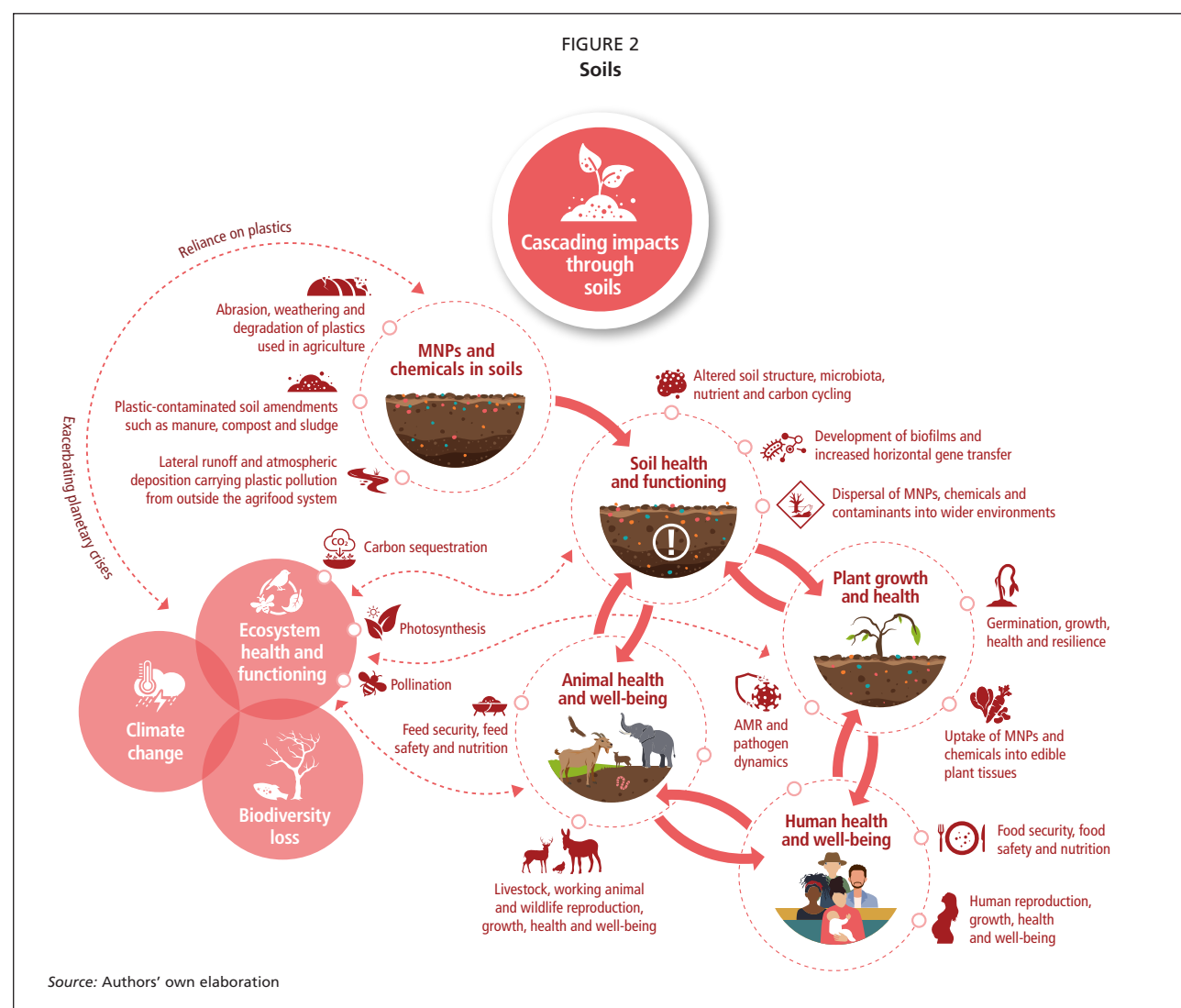
*Trace the cascading effects of plastic pollution from across the life cycle and its effects on agrifood systems and One Health. Open any pathway to see how the impacts reverberate across populations and environments.*

### 3. Illustrative impact pathways of plastic pollution and One Health

The following section presents the impact pathways illustrating how plastic pollution moves through agrifood systems, creating interconnected pathways of exposure and effects across animal, human, and environmental health. While presented separately for clarity, these pathways are closely interconnected.

Plant health is embedded throughout the illustrative impact pathways, particularly within the soil pathway, reflecting the central role of soils in supporting crop production and transmitting contaminants through agrifood systems

#### IMPACT PATHWAY 1: CASCADING IMPACTS THROUGH SOILS



Soils (see [Figure 2](#)) are fundamental to sustaining all life on Earth and are therefore central to One Health ([Banerjee & van der Heijden, 2023](#)). They support plant and animal health, regulate water and climate, and underpin food security, nutrition and food system resilience.

Plastic pollution is accumulating rapidly in soils globally, from sources both inside and outside of agrifood systems. The use of plastic agricultural films, mulches and polymer-coated fertilizers are direct contributors to soil pollution alongside plastics-contaminated soil amendments such as manure, compost and sludge ([Oliveri Conti et al., 2024](#); [Nizzetto et al., 2025](#); [Cusworth et al., 2024](#); [Duquesne et al., 2025](#); [Colombini et al., 2024](#)). Mechanical abrasion and weathering of plastic materials during and after use, poor recovery rates, and open burning of plastics waste, release MNPs and toxic chemicals into the air and soils ([Huang et al., 2020](#); [Nizzetto et al., 2025](#)). Lateral runoff and atmospheric deposition pollute soils with plastics from multiple sources beyond farm environments, including tyre particles and textiles ([UNEP, 2021](#)).

Over half of the studied agricultural soils worldwide already show MNP concentrations at detected levels of up to 0.3 percent ([Nizzetto et al., 2025](#)). MNP concentrations as low as 0.001 to 0.05 percent have been shown to impair soil structure, disrupt soil microbiota and invertebrate's metabolism, and affect nutrient and carbon cycling ([Chaudhary et al., 2025](#)). Smaller particles have a larger relative surface area from which hazardous chemicals present in plastics can leach into soils ([Teuten et al., 2009](#)). In addition, MNPs can adsorb environmental contaminants such as heavy metals and persistent organic pollutants (POPs), potentially influencing their environmental fate, mobility and bioavailability and contributing to complex contaminant mixtures in agricultural soils ([Fred-Ahmadu et al., 2020](#)). MNPs can also act as microbial attachment surfaces, encouraging the development of biofilms and horizontal gene transfer between bacterial

communities, which can alter soil microbiomes and potentially exacerbate AMR ([Yu et al., 2023](#)). MNPs and the chemicals, contaminants and bacteria they may carry, are highly mobile and can disperse to surrounding environments and waterways through rainfall and surface run-off.

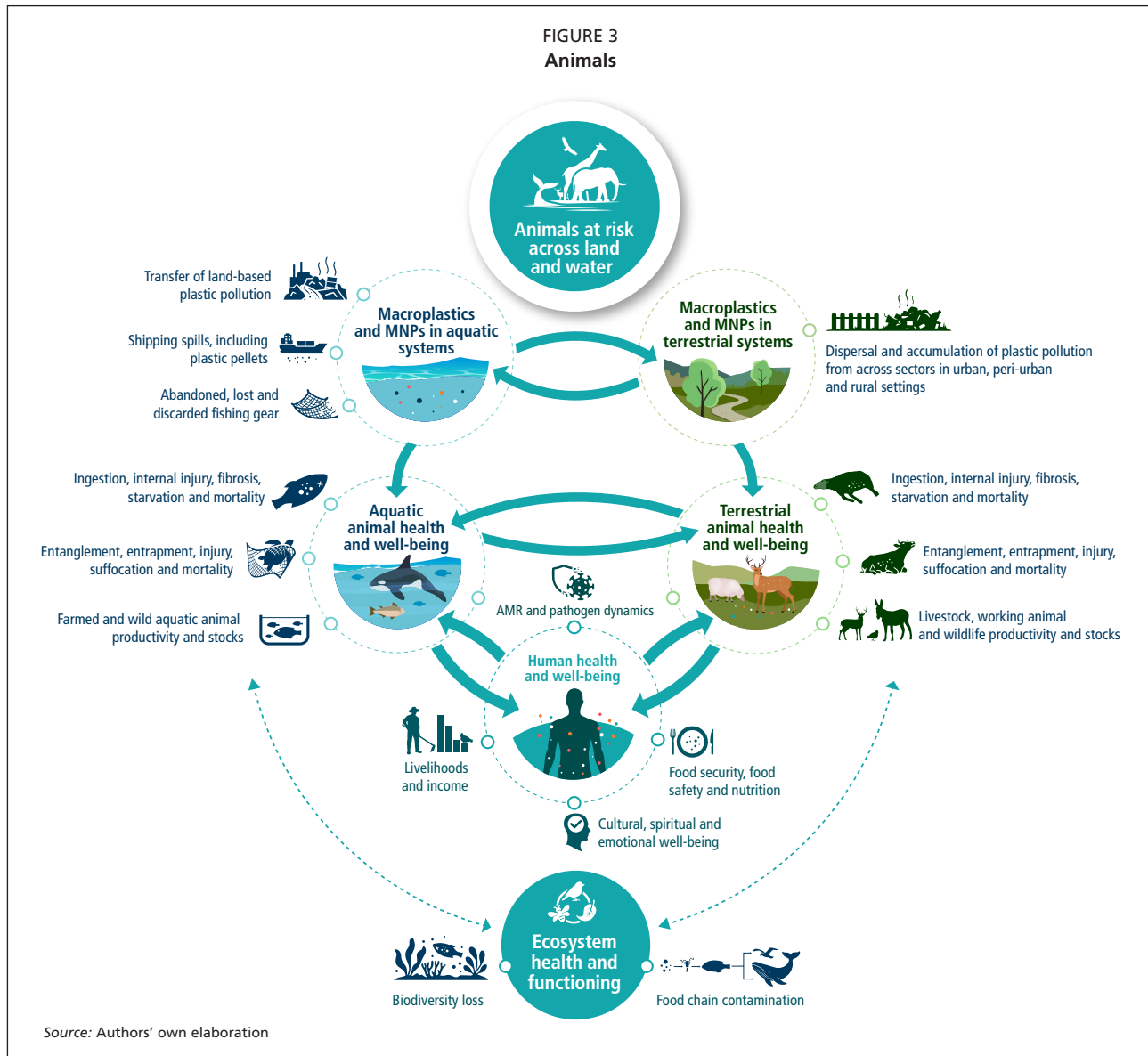
These pathways multiply the hazards of MNPs and other pollutants alike, creating new compounding mixture effects on soil functioning and plant health ([Qin et al., 2025](#)).

Documented adverse effects of plastic pollution on plants include inhibited seed germination, seedling establishment, and reduced root/shoot growth, with potential consequences for biomass and yield ([Nizzetto et al., 2025](#); [Chaudhary et al., 2025](#)). MNPs can obstruct vascular tissues, trigger oxidative stress that damage cell membranes, reduce chlorophyll content, cause DNA damage and altered gene expression, impairing development and stress responses ([Nizzetto et al., 2025](#); [Chaudhary et al., 2025](#)). Plants can take up MNPs and plastic chemicals directly from soils through their roots, including into edible areas destined for human and animal consumption ([Oliveri Conti et al., 2020](#); [Nizzetto et al., 2025](#)).

Emerging evidence linking the soil-plant-animal/human microbiome suggests that these dietary pathways therefore present new risks for AMR and pathogen dynamics ([Banerjee & van der Heijden, 2022](#); [Ma et al., 2025](#); [Zhao et al., 2025](#)).

Together, these effects of MNP soil pollution may be disrupting critical ecosystem services such as pollination ([Sheng et al., 2024](#)), photosynthesis ([Zhu et al., 2025a](#)), and carbon sequestration ([Zhu et al., 2025b](#)). Increasing levels of plastic pollution in soils presents potentially irreversible long-term risks for ecosystems, agricultural productivity, animal health, human food security and nutrition, and threatens to lock people and planet into a cycle of reliance on plastics with cascading impacts across One Health ([Gencer et al., 2026](#); [Sheng et al., 2024](#)). In turn, this exacerbates planetary crises including climate change and biodiversity loss ([Villarrubia-Gómez et al., 2024](#)).

## IMPACT PATHWAY 2: ANIMALS AT RISK ACROSS LAND AND WATER



Plastic ingestion and entanglement are widespread among animals across aquatic and terrestrial systems, from microfauna, to insects including pollinators and other key species, to the largest animals on the planet (Daghighi et al., 2023; Sheng et al., 2024; Kahane-Rapport et al., 2022). These interactions have been documented for decades (Kühn & van Franeker, 2020; Santos et al. 2021), and rising awareness of One Health interlinkages now underscores the direct onwards risks to human food security, nutrition and livelihoods, as shown in Figure 3 above.

Aquatic environments are major sinks of plastic pollution globally, with up to 23 million tonnes entering rivers, lakes and oceans annually (Borrelle et al., 2020). While abandoned, lost, and discarded fishing gear (ALDFG) represent major components of marine plastic pollution, the largest flows of

plastic pollution into these environments come from land-based sources transported through stormwater runoff, lakes, rivers and wind dispersal (Carney Almroth & Eggert, 2019). Among these are many agrifood system applications such as agricultural plastics, food and beverage packaging, and tyre particles (Morales-Caselles et al., 2025; Kelly et al., 2026; Sun et al., 2025).

The accumulation, weathering, and fragmentation of plastics, and their potential for long-range mobility in aquatic environments, result in widespread exposure to hazardous items, plastics particles and associated chemicals across freshwater and marine species (Santos et al., 2021; Gündoğdu et al., 2024; Thompson et al., 2024).

Ingestion of plastic pollution is widely documented among animals in aquatic and terrestrial environments, who may

consume plastic items, fragments and particles by accident or mistake them for food. Microplastics have been detected in more than 1 300 species, including fish, mammals, birds, and insects (Murphy et al., 2025; Santos et al., 2021; Kuhn, 2020; Edo et al., 2021). For example, it is estimated that some blue whales may ingest 10 million pieces of microplastic per day, the approximate equivalent of 43.6 kg (Kahane-Rapport et al., 2022). As plastic pollution continues to fragment into smaller particles in the environment it becomes bioavailable to a more extensive range of organisms which in turn increases the risk of widespread food chain contamination. Ingestion of plastic debris can lead to internal injury, fibrosis, starvation and mortality, including via the leaching of toxic chemicals and adsorbed pollutants (Tian et al., 2021; Stevens et al., 2024).

Compounding this impact pathway, larger fish and marine animals such as whales, dolphins and turtles are highly susceptible to entanglement in macro-plastics, including abandoned, lost or otherwise discarded fishing gear, leaving them trapped, injured or suffocated, often leading to injury or death (Gall & Thompson, 2015).

Together, the effects of plastic ingestion and entanglement in aquatic environments may place delicate food webs, ecosystems and aquatic stocks at risk of irreversible harm (Beaumont et al., 2019; MacLeod et al., 2021). For the 61 million people engaged in fisheries and aquaculture worldwide, the 3 billion people relying on aquatic animal foods as a primary source of protein (FAO, 2024a), or the Indigenous Peoples whose diets and cultures are increasingly eroded by plastic pollution impacts, this represents a serious threat to livelihoods, incomes, food security and safety, nutrition, health and spiritual and emotional well-being (Yose et al., 2023; Beaumont et al., 2019).

On land, particularly across urban, peri-urban and rural settings with limited waste management infrastructure, plastic debris accumulates across roadsides, rural landscapes and smallholder farms (Mihai et al., 2022). Ingestion of this plastic is common among grazing ruminants, poultry and working animals (Haddy et al., 2025; Priyanka & Dey, 2018). The adverse consequences and complications can be both chronic and acute, ranging from suffocation, impaction and malnutrition to immobility, morbidity and death (Priyanka & Dey, 2018). As a result of dietary and environmental exposure of livestock, terrestrial animal products are often contaminated with microplastics (Prata & Dias-Pereira, 2023).

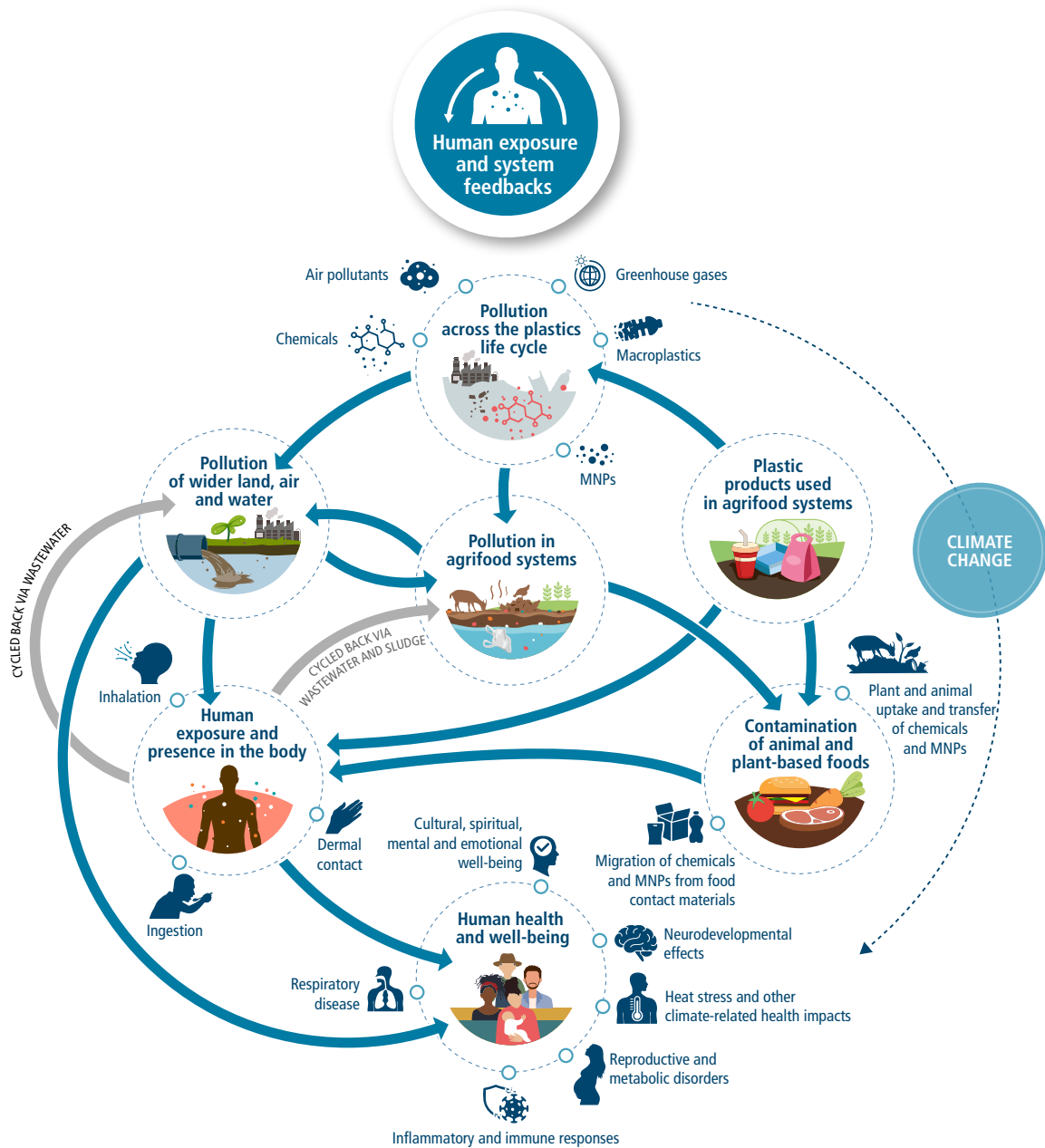
Working equines and livestock provide a particularly illustrative example of vulnerabilities to plastic ingestion. In areas with limited or insufficient waste management, free-roaming equines and livestock frequently ingest plastic debris, resulting in gastrointestinal obstruction, poor body condition, reduced welfare and, in severe cases, death. These impacts extend beyond animal health, affecting the livelihoods of communities that depend on working animals for transport and agricultural activities (Haddy et al., 2025). Microplastics can also affect ruminal fermentation, including protein metabolism (Tassone et al., 2025). They have also been shown to reduce growth performance and alter gut microbiota in poultry, with exposed animals requiring the same amount of feed while gaining less weight (Zou et al., 2023).

Plastic pollution can create conditions that favour the growth and exchange of pathogens and AMR genes. Microplastics provide surfaces for microbial colonization, forming “plastispheres” where bacteria, including potentially pathogenic and resistant strains, can persist and interact (Yang et al., 2025a; Yang et al., 2025b). Animals that ingest plastics or are otherwise exposed to these plastispheres may face increased risks of treatment-resistant infections, with potential implications for zoonotic disease transmission and broader AMR risks to human health (Stevenson et al., 2025; Uluseker et al., 2021). Addressing these infections can lead to higher veterinary and economic costs and may drive increased antibiotic use, further exacerbating AMR (WOAH, 2024).

In parallel, conventional wastewater treatment processes are not designed to fully remove antibiotic-resistant bacteria or resistance genes. As a result, treated effluent released into the environment can continue to interact with microplastics, creating additional pathways for the formation and spread of plastispheres enriched with pathogens, AMR bacteria and genes (Uluseker et al., 2021).

Pastoralists who play a key role in community nutrition, and smallholders who produce around one-third of the world’s food supply, often depend heavily on livestock and working animals (Lowder et al., 2021). These people are also among the most climate-vulnerable populations on Earth and are therefore disproportionately exposed to the cascading effects of plastic pollution on animal health, livelihoods, food security, nutrition and their own well-being (IPCC, 2022a; Niles and Salerno, 2018). These impacts are not confined to the most vulnerable populations, but reverberate across agrifood systems, economies and societies worldwide.

## IMPACT PATHWAY 3: HUMAN EXPOSURE AND SYSTEM FEEDBACKS

FIGURE 4  
Humans

Source: Authors' own elaboration

Plastics used in agrifood systems release chemicals and MNPs throughout their normal and intended use, and across their full life cycles from extraction of their raw materials to eventual waste and widespread dispersal in the natural environment (Zimmermann et al., 2021; Zimmermann et al., 2025). Plastic pollution from other sectors, including via extraction of raw materials to eventual waste and widespread dispersal in the natural environment, is also transmitted through agrifood systems which act as a major route of human

exposure to chemicals associated with plastics and MNPs. As shown in Figure 4 above, these exposures happen including via ingestion, inhalation and dermal contact (Thompson et al., 2024). This includes occupational exposures among those directly involved in handling plastics such as manufacturers, farmers, food processors and informal waste pickers (ILO, 2023); as well as chronic population-level dietary exposures via contaminated foods (Geueke et al., 2024).

More than 16 000 chemicals have been found in plastics, most of which are unregulated globally ([Monclús et al., 2025](#)). A quarter of these chemicals are already considered hazardous due to their persistence, bioaccumulation, mobility, and/or toxicity and 66 percent lack official hazard classifications by regulatory agencies or industry ([Monclús et al., 2025](#)). From food production to processing, retail, packaging and consumption, plastics are in direct contact with food and beverages, often under conditions that promote chemical migration into products destined for human diets and nutrition ([Muncke et al., 2025](#)). Migration may also involve contaminants introduced during reuse or recycling processes ([Carney Almroth et al., 2025](#); [Geueke et al., 2023](#); [FAO, 2024c](#)), and chemicals released from bio-based plastics and other alternative materials ([Monclús et al., 2025](#); [Zimmermann et al., 2020](#)).

A quarter of the 14 000 chemicals found in plastics and other food contact materials, have also been identified in human bodies, including known endocrine disruptors such as bisphenols and phthalates ([Geueke et al., 2024](#)). Chemicals in plastics have been linked to effects on reproductive health, neurodevelopmental and behavioural disorders, obesity, and changes in hormone levels, for some at extremely low doses, and are particularly concerning during critical windows of development, including throughout pregnancies and childhood ([Maffini et al. 2021](#); [Symeonides et al., 2024](#)).

Modern industrial diets are increasing human exposure to these chemicals including through ultra-processed foods (UPFs), which represent over 50 percent of calorie intake in many countries and are heavily reliant on plastics ([Monteiro et al., 2025](#); [Yates et al., 2024](#); [Ralston et al., 2026](#)). Evidence indicates that diets high in UPFs, notably among population sub-groups such as women, children and Indigenous Peoples, are associated with higher presence of MNPs and plastic chemicals ([Milne et al., 2024](#); [Preston et al., 2025](#)).

At the same time, diets of fruit and vegetables even without plastic packaging also contain bisphenols and phthalates, suggesting that agricultural production environments are contaminated with chemicals including those associated with plastics ([Harvey et al., 2026](#); [Chen & Hofmann, 2026](#)).

MNPs have been identified in several food groups, with evidence demonstrating their transfer from food contact materials ([Zimmermann et al. 2025](#)) among other environmental sources ([Thompson et al. 2024](#)).

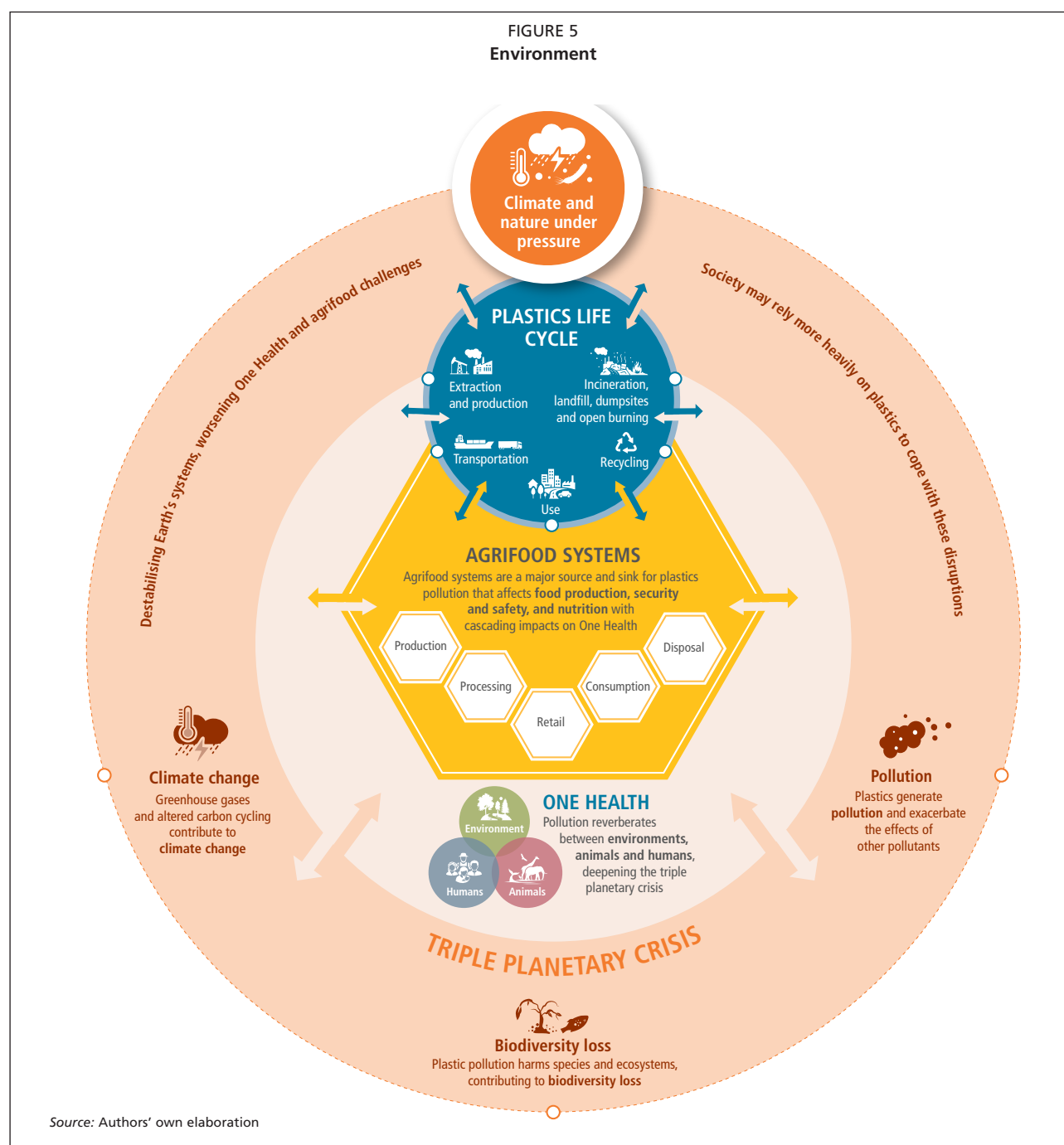
MNPs have been identified in placenta, breast milk, testes, stools, and other human tissues ([Thompson et al. 2024](#)). Human exposure therefore now begins in utero and continues across the life course ([Landrigan et al., 2025](#); [Symeonides et al., 2024](#)). Exposure to MNPs may cause damage because of their physical presence and the release of the chemicals plastics contain ([Wehrli et al. 2025](#)). Animal models and emerging evidence in human tissues suggests MNPs may be linked to gut lining damage, altered microbiomes, inflammatory and immune responses, and reproductive toxicity, with the smallest particles potentially crossing biological barriers and reaching vital organs ([Pulvirenti, et al., 2022](#); [Prata 2023](#), [Liebgott et al., 2026](#); [Wehrli et al., 2025](#)). This includes evidence showing that the presence of MNPs may exacerbate liver disease ([Chokshi et al., 2026](#)).

Uncertainties remain regarding the health and food safety implications of current dietary exposure levels, particularly as many experimental studies rely on commercially available model particles and controlled exposure conditions that differ from environmentally weathered MNPs ([Gouin et al., 2024](#)). Nevertheless, the evidence of harms is growing alongside advancing harmonized methods for detecting MNPs, and understanding their fate and impacts in living organisms.

MNPs are discarded or excreted and accumulate in wastewater and sewage sludge, which are widely applied to farm soils as fertilizers, thus creating a feedback loop that affects plants, animals and natural environments ([Weithmann et al., 2018](#); [Rabbi et al., 2026](#)). This multiplies risks for food production, food security and food safety.

Human exposure to plastics is compounded by health hazards across the life course but also across the plastics life cycle. Before plastics used in agrifood systems even reach their intended use, extraction of raw materials, manufacturing, transport, and production of plastics release toxic emissions that disproportionately affect nearby communities ([UNEP, 2011](#), [UNEP, 2021](#); [Johnston & Cushing, 2020](#)), as well as result in GHG emissions that contribute to climate change and its impacts with knock-on effects across the entire food value chain ([Deeney et al., 2026](#); [Karali et al., 2024](#); [Villarrubia-Gómez et al., 2024](#)). The chronic and widespread nature of plastic pollution is placing a continuous burden on public health and well-being while simultaneously cycling back through the ecosystems, plants and animals that underpin One Health.

## IMPACT PATHWAY 4: CLIMATE AND NATURE UNDER PRESSURE



Pollution across the entire plastics life cycle from fossil fuel extraction, production, transport, use and end-of-life is rapidly growing and exacerbating impacts across Earth's system processes upon which agrifood systems depend (Villarrubia-Gómez et al., 2024).

Plastic production accounted for 5.4 percent of GHG emissions in 2019 and is projected to triple by 2060 (Karali et al., 2024), placing plastics firmly within the climate crisis. Open burning of plastics is a neglected yet critical source of

GHGs (IPCC, 2022b; Pottinger et al., 2024). At the same time, carbon cycling and sequestration is undermined by land-use change linked to fossil fuel extraction and other plastic life-cycle activities as well as the accumulation of MNPs in soils and aquatic environments (Zhu et al., 2025b). Together, these processes intensify climate change, environmental stress, and agrifood-system vulnerability, creating a feedback loop that may increase reliance on plastics for climate adaptation, including protective structures, irrigation, row covers, and shelf-stable packaged food and water, as shown in

**Figure 5** above (FAO, 2021; Wang et al., 2025). This expansion of plastics use also contributes to the eroding of Indigenous People's food systems, agroecological and conservation knowledge systems that have pre-dated the introduction and proliferation of these materials (Yose et al., 2023; Aswani et al., 2018).

Plastics exacerbate different forms of pollution, beginning with emissions from fossil fuel extraction, spills and leaks from pipelines and transportation, and pollution from petrochemical refining and plastic production. In addition, MNPs across terrestrial and aquatic environments provide optimal surfaces for colonization by diverse microorganisms (Zettler et al., 2020). This ever-growing 'plastisphere' facilitates the propagation of biofilms, horizontal gene transfer and other pathogens, transporting them across long distances and into contact with new agroecological systems, animals and humans, hence increasing global risks for food safety, AMR and infectious disease (Stevenson et al., 2025). Moreover, when plastic breaks down into MNPs, the ratio of surface area to volume increases (Rahman et al., 2024). As plastic fragments become smaller through weathering, their relative surface area increases, creating more surface from which hazardous chemicals in plastics may leach into surrounding soils (Teuten et al., 2009).

This dramatic increase in available surface area provides greater opportunities for microbial colonization and biofilm formation. After use, agrifood-system plastics are often weathered, contaminated, or are too structurally complex to recover and recycle effectively (Hofmann et al., 2023; Ada et al., 2023). Plastics are therefore frequently incinerated, landfilled, or openly burned, creating new exposure pathways for humans, other animals and the

environment, and contributing to persistently low global plastics recycling rates of about 9 percent (Houssini et al., 2025). Plastic recycling facilities can themselves become significant point sources of microplastic pollution when wastewater is inadequately treated or illegally discharged, a risk that is amplified in countries that are major recipients of the international plastic waste trade (Gündoğdu & Avcıoğlu, 2026).

While the full effects of plastics on biodiversity have not yet been quantified, key species and ecosystem services essential for agrifood system functioning are exhibiting evidential stress from plastic pollution and the microbial biotransformation it is catalysing.

For instance, sublethal effects i.e. those that are not immediately fatal, but still cause harm to an organism's health, behaviour, or ability to survive and reproduce, caused by plastic pollution have been observed among beneficial insects such as pollinators and bio-control agents such as natural pests (Sheng et al., 2024). Research suggests that this may already be undermining plant-pollinator interactions upon which over two-thirds of global crops depend (Klein et al., 2007; Aizen et al., 2019; Shah et al., 2023). Meanwhile, a plethora of invertebrates that support terrestrial and aquatic ecosystem services such as water filtration and retention, and nutrient cycling, are also showing signs of strain (Silva et al., 2022). The agronomic implications of such strain could unravel a range of new harms including impaired biocontrol, greater use of agrochemicals and their associated effects.

In summary, plastic pollution plays a major role in the triple planetary crisis of climate change, pollution, and biodiversity loss that threatens One Health (Schmidt et al., 2024; Villarrubia-Gómez et al., 2024; Winkler et al., 2025).



## 4. Aligning evidence with policy: an integrated One Health approach

Addressing plastic pollution in agrifood systems effectively requires a shift from fragmented, end-of-life responses toward a One Health approach that recognizes the interdependence of ecosystems, production systems, and societies across the full life cycle of plastics. The four policy pillars that follow translate this perspective into action. Together, they call for framing the problem systemically through integrated evidence and data; co-designing inclusive, equitable solutions grounded in real-world contexts; strengthening coordinated governance and

joint action across sectors and scales; and institutionalizing continuous learning through participatory review and living evidence.

Taken as a whole, these four pillars provide a practical foundation for a governance framework that is technically robust, socially equitable, adaptable over time, and coherent with global environmental and health commitments to address plastic pollution in agrifood systems.



### PILLAR 1: A SHARED EVIDENCE BASE CONNECTED ACROSS SECTORS

Plastic pollution in agrifood systems is often governed through fragmented institutional mandates (FAO, 2026) and inadequate data and monitoring systems (Yates et al., 2025) that fail to capture what plastics are used across global agrifood systems, how they move through soils, water, biota and food webs, and how they interact with biodiversity loss, chemicals and AMR. A shared evidence base therefore requires breaking down institutional and disciplinary silos and building a shared understanding of the potential hazards and risks plastic pollution poses across agriculture, aquaculture, environment, health, fisheries, water and waste sectors.

National One Health platforms and policy frameworks, for instance, could support this shift by promoting science-policy interfaces and interoperable surveillance, monitoring and assessment systems that integrate environmental, food safety, veterinary and public health information alongside the knowledge and full and effective participation of Indigenous

Peoples. Such approaches can help assess risks and identify cumulative impacts, emerging hazards, and feedback loops. Harmonized metrics, integrated data governance and “One Health intelligence” can further strengthen cross-sectoral decision-making (FAO, UNEP, WHO, and WOA, 2023).

A systems perspective is equally important for policy design. Measures targeting a single stage of the plastics life cycle or a single subsector of the agrifood system may shift harms elsewhere in the system or generate unintended consequences for other populations. Drawing together One Health and agrifood systems transformation approaches will ensure that the precautionary principle can be operationalized for effective action and policy that mitigate burden shifting and maximise co-benefits (FAO, 2025b). Applying the precautionary principle can help to proactively identify trade-offs, avoid burden shifting and maximize co-benefits across multiple policy objectives.



### PILLAR 2: INCLUSIVE SOLUTIONS CO-DESIGNED WITH COMMUNITIES

Technical solutions and material substitutions alone will not deliver a just transition away from harmful plastic products, especially if they disregard social realities, power imbalances and local livelihoods. Actions towards ending plastic pollution must be “socially anchored”, and systematically aligned with agroecological transitions towards more healthy, resilient and equitable food systems (HLPE, 2019; FAO, 2022b). Within this context, the emerging bioeconomy, particularly when grounded in sustainability principles, can support transitions away from plastics, that include sustainable resource use and innovation in materials and systems where necessary, but must be guided to ensure that it contributes to, rather

than undermines, food security, ecosystem integrity and social equity (FAO, 2024b).

These efforts should draw upon diverse disciplines, including humanities, social sciences and the sectoral and local expertise of workers and communities to ensure that these responses are effective, meaningful, equitable, acceptable and that any unintended consequences or “burden-shifting” are mitigated. Actions should be based on a comprehensive approach that addresses the full life cycle of plastics used in agriculture, including upstream interventions focusing on prevention or reduction (FAO, 2025a), informed by evidence and developed through co-production models that formally

include smallholder farmers, waste pickers, Indigenous Peoples, artisanal fishers, women's groups and other populations disproportionately affected by decisions on which materials and practices are reduced, removed or replaced. This principle of co-design can improve the real-world feasibility, legitimacy and uptake of removing unnecessary plastics from across diverse agrifood systems and finding safer and more sustainable alternative systems, processes, practices or materials where needed to maintain essential functions.

Impact pathway analyses of agrifood systems transitions consistently show that poorly designed top-down measures can raise production costs, compress margins for smallholders and entrench existing inequalities (Arata et al., 2025; Carlile et al., 2026). To avoid this, stricter standards on agrifood system plastics production, transport, use and lifecycle management (for example, product stewardship, recyclability and robust toxicity testing, including mandatory measures) could be coupled with targeted subsidies, concessional credit and technical assistance for small-scale producers and micro-enterprises. Incentivizing transitions away from petrochemicals and plastics and promoting safe-by-design practices along the entire agrifood value chain from fertilizer, feed packaging, irrigation greenhouse infrastructures, to processing materials and packaging could result in minimization of leakage and harmful chemicals from the outset rather than managed as end-of-life plastic waste.

Bio-based and/or biodegradable plastics may be able to play a role in reducing dependence on fossil-based polymers and associated emissions, however, evidence indicates that they are not yet demonstrably better for the environment in

terms of toxicity and degradation (Shanmugam et al., 2026).

Bio-based materials are also not necessarily biodegradable, and biodegradable materials are not necessarily bio-based (Scientists Coalition, 2023; Gündoğdu et al., 2026). In addition to feedstock-related natural toxins and allergens (Lacourt et al., 2024, FAO, 2022b) they contain hazardous synthetic chemicals that are similar and/or exceed conventional agrifood system plastics (Zimmermann et al., 2020; Alcaraz-Dólera et al., 2026). Large-scale shifts in feedstock demand within bioeconomy may have land use, water resource, and food security implications (Brizga et al., 2020; Ahmed et al., 2021; FAO, 2024b).

If used, the production of bio-based feedstocks should prioritize residues, wastes and other non-food biomass, operate within local land and water constraints, and safeguard food security by avoiding competition for productive land and the displacement of smallholders. In addition, robust toxicological and environmental assessment of alternative materials, including bio-based and biodegradable materials, is essential to ensure they do not introduce new, poorly understood contaminants into the soil–water–food nexus (Lacourt et al., 2024, FAO, 2022b). Critically, claims of “biodegradability” must be matched by performance under realistic agronomic conditions (Alcaraz-Dólera et al., 2026; Gündoğdu et al., 2026). Food safety safeguards should be an integral part of circular approaches to ensure that contaminants, including legacy substances and other hazardous chemicals, are not unintentionally reintroduced into food-contact materials or applications (FAO, 2024c).



### PILLAR 3: COORDINATED ACTION ALIGNED ACROSS POLICIES AND GLOBAL AGREEMENTS

Scientific evidence can only translate into action when supported by clear institutional coordination. Siloed mandates across ministries of agriculture, environment and water, labour, health, and trade remain a primary barrier to comprehensive plastics governance across agrifood systems. Countries could establish or strengthen high-level national One Health platforms with explicit responsibility for plastics and plastic pollution in agrifood systems that bridge key ministries and agencies. Such platforms could have advisory authority to set joint, binding targets for reducing plastic use and leakage across agrifood systems, improving environmental quality indicators and monitoring (e.g., soil and water pollution thresholds) and tightening national food-safety standards related to microplastics and plastics-associated chemicals. Their work programmes should include regular assessments, stakeholder consultations, independent science-policy interfaces, and the co-development and sharing of guidance and tools tailored to sectoral ministries and subnational authorities. Such platforms can also coordinate investments, training and communication strategies, ensuring that

subnational authorities and communities are also supported in implementation.

At national levels, collaboration across multiple ministries can accelerate One Health policies on plastic pollution in agrifood systems, while an effective global plastics treaty could facilitate international harmonisation. Alignment is also essential with the Kunming–Montreal Global Biodiversity Framework, particularly Target 7 on reducing pollution (including plastic pollution) to levels not harmful to biodiversity and Target 10 on sustainable management of agriculture, aquaculture, fisheries and forestry. Action on plastic pollution could likewise be integrated into implementation of the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement, including through nationally determined contributions (NDCs), adaptation planning and circular economy measures that reduce GHG emissions across the plastics life cycle while strengthening resilience in agrifood systems. In parallel, national frameworks should build on and complement the Basel Convention, Rotterdam Convention and Stockholm Convention (BRS Conventions), which provide

internationally agreed mechanisms for the environmentally sound management of plastic wastes, hazardous chemicals and persistent organic pollutants (POPs) throughout the plastics value chain. Coherent implementation across these multilateral environmental agreements can improve policy consistency, reduce duplication and provide regulatory certainty, enabling investment in circular business models, retrieval systems, extended producer responsibility (EPR), safer materials and monitoring innovation ([UNECE, 2025](#)).

Sustaining such interfaces also requires predictable financing. Domestic budget lines complemented where appropriate by international financial mechanisms, could cover core functions such as integrated monitoring, assessment cycles, stakeholder engagement, and capacity building. Multilateral development banks and climate and biodiversity funds can play an important role in de-risking investments in safer systems, practices and alternatives that emerge from these science-based processes.

Recent debates around the design of the UN Intergovernmental Science-Policy Panel on Chemicals, Waste and Pollution Prevention (ISP-CWP), as well as around the global plastics treaty negotiations, and its future science-policy interface underline the importance of safeguarding independent scientific input from undue influence by vested interests ([Schäffer et al., 2023](#)). National One Health platforms and related bodies should adopt safeguards, ensuring that policy advice is based on a balanced consideration of peer-reviewed evidence, Indigenous Peoples' food and knowledge systems, and real-world implementation

experience. Consistent with the principles of the [United Nations Declaration on the Rights of Indigenous Peoples \(UNDRIP\)](#), mechanisms should also be established to enable the systematic and meaningful participation of Indigenous Peoples in decision-making processes, including the development, assessment and implementation of policies that may affect their lands, territories, resources, food systems, livelihoods and knowledge systems.

Legal and institutional arrangements should embed “living evidence” mechanisms whereby continuous field monitoring of plastics contamination, AMR markers and ecosystem health indicators is formally linked to regulatory review cycles that draw on the latest independent science. For example, pre-agreed triggers such as detection of concerning levels of microplastics in feed and foods, new evidence of harmful chemical additives, or rapid shifts in AMR patterns associated with environments heavily affected by plastic pollution could automatically initiate structured policy updates or emergency measures.

By institutionalizing such feedback loops, governments can maintain precaution, while adapting regulations as evidence evolves, rather than waiting for crises to force abrupt and disruptive policy shifts. Sub-national authorities often responsible for land-use planning, waste management and environmental regulation should be fully integrated into these feedback loops, with tailored guidance and resources to adapt national frameworks to local ecological and socio-economic conditions.



## PILLAR 4: RESPONSIVE REVIEW INFORMED BY LIVING EVIDENCE

Building on the monitoring, evaluation and learning (MEL) concepts set out in the implementation and monitoring guidance of the One Health Joint Plan of Action (OH JPA), training, fellowships, and peer-to-peer exchanges, can help build a critical mass of expertise to sustain iterative assessment and policy revision. Continuous, community-engaged monitoring could capture not only pollution levels but also local perceptions of risk, livelihood impacts and the performance of alternatives.

Scheduled policy reviews and horizon scanning (for example, every three to five years) by multi-stakeholder bodies protected from conflicts of interest could examine evidence on pressures (extraction, transportation and refining of feedstocks, plastic pollution and use), state (contamination levels), exposure (in humans, animals and ecosystems), outcomes (health and productivity) and equity (who benefits, who bears the burden), and then adjust regulations, incentives and support schemes accordingly.

Pillar 4 ensures that One Health governance of plastics remains responsive, inclusive and aligned with broader goals on climate, biodiversity, food security, food safety and social justice.

A formal, human rights-based approach centred explicitly on the right to food, the right to health, and the right to a clean, healthy, and sustainable environment provides a binding normative anchor for all science-to-policy processes on plastics use in agrifood systems. The recent UN recognition of the right to a clean, healthy and sustainable environment as a universal human right reinforces the obligation of States to prevent plastic pollution from undermining the conditions for health, food and ecological integrity, and provides an additional legal basis for precautionary and equity focused action on plastics used in agriculture and across food systems ([A/RES/76/300](#)). Equally, the UNFCCC and related international climate laws impose legal obligations on states to address GHG emissions, as recently clarified by the International Court of Justice (ICJ) in its Advisory Opinion on Obligations of States in Respect of Climate Change ([ICJ Advisory Opinion](#)).

These pillars treat plastics not merely as an isolated waste product to be managed after the fact, but as a cross-cutting determinant of human, animal, plant, and ecosystem health that must be governed proactively and coherently across its entire life cycle, and across all sectors, populations and scales.

# Conclusion

Plastic pollution in agrifood systems threatens the health of soils, plants, animals, humans and broader environmental systems that together constitute One Health. The evidence summarized in this technical brief illustrates how plastics pollution and harmful emissions, including macroplastics, MNPs and chemicals, air pollutants and GHGs, are released across plastics life cycles, including from plastics intentionally used in agrifood systems. Plastic particles and chemicals contaminate and move through agrifood systems, simultaneously facilitating the spread of other pollutants, bacteria and pathogens that may exacerbate AMR, while air pollutants and GHGs exert wider pressures on climate and environmental health. Together, these forms of pollution create multiple pathways of interconnected, compounding and cascading impacts across agrifood systems and One Health.

The illustrative impact pathways presented in this brief highlight four diverse examples of how the effects of plastics pollution on soils, animals, humans and Earth's biophysical systems reverberate across broader One Health interlinkages in ways that collectively undermine food production and safety, environmental and planetary health, ecosystems and biodiversity, and human health and well-being. These impacts are particularly severe for populations whose livelihoods, food security, health and cultural, spiritual and emotional well-being are most directly affected by plastic pollution.

Nevertheless, given the fundamental role of agrifood systems in human existence, and the fact that these systems are both a major source and sink of plastic pollution, all populations are at growing risk.

A One Health approach recognizes that adverse effects of plastic pollution do not occur in isolation. Connecting evidence across human, animal, plant and environmental health reveals how plastics pollution and its associated impacts move between organisms and systems, where impacts interact or accumulate, and where interventions can prevent or reduce harms before they cascade. It can also help identify trade-offs and avoid actions in one part of the system that inadvertently shift harms elsewhere.

This technical brief contributes to the evolving FAO One Health in Agrifood Systems Framework by illustrating how a One Health lens can be applied to understand the cascading effects of plastic pollution. Translating this approach into action requires stronger and more connected monitoring and assessment; context-specific and socially grounded solutions; coordination across sectors, disciplines and mandates; and adaptive policies capable of responding to emerging evidence. Together, these elements can support a shift from fragmented responses to more preventive, integrated action across the plastics life cycle and agrifood systems.



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